Spatial–Temporal Collaborative Network for Satellite Image Time Series Semantic Change Detection

Jiangwei Xie, Yuxiang Zhang, Mengmeng Zhang, *Member, IEEE*, Yan Du, and Wei Li, *Senior Member, IEEE*

***Abstract*—Semantic change detection (SCD) aims to generate both semantic segmentation maps and binary change (BC) maps from binary temporal remote sensing images and plays a crucial role in remote sensing image interpretation. While learning-based methods have shown strong performance on standard benchmarks, the more challenging satellite image time series (SITS)-based SCD task remains underexplored. Existing SITS-SCD approaches often overemphasize temporal cues while neglecting effective spatial–temporal interaction and are stuck in low-efficiency feature fusion, limiting their representation capacity. To this end, we propose ST3DUNet, a spatial–temporal collaborative network for SITS-SCD. ST3DUNet adopts a U-shaped architecture that extracts hierarchical spatial–temporal representations through a bidirectional spatial–temporal module (BSTM) and recursively fuses them via a collaborative fusion block (CFB) based on dynamically updated prototype matrices. This collaborative fusion enhances decoding by leveraging informative spatial–temporal embeddings. Moreover, we evaluate the robustness of our model under spatial domain shifts, which adhere to real-world remote sensing image decryption conditions. Extensive experiments on three SITS datasets demonstrate that ST3DUNet consistently outperforms independent temporal processing, pairwise temporal modeling, and existing SITS-SCD methods, as well as three Earth observation foundation models (FMs), achieving consistent improvements of approximately +5%, +3%, and +5% in mean intersection-over-union (mIoU) across DynamicEarthNet-RGB, DynamicEarthNet-Sentinel2, and Muds, respectively. Code will be released at https://github.com/xuwayyy/ST3DUNet**

***Index Terms*—Bidirectional spatial–temporal module (BSTM), collaborative fusion block (CFB), satellite image time series (SITS), semantic change detection (SCD).**

Received 29 April 2026; revised 3 June 2026; accepted 9 July 2026. Date of publication 14 July 2026; date of current version 27 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62471040 and Grant W2411055. *(Corresponding author: Mengmeng Zhang.)*

Jiangwei Xie, Mengmeng Zhang, and Wei Li are with the School of Information and Electronics and the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing, Beijing Institute of Technology, Beijing 100081, China, and also with Beijing Institute of Technology, Zhuhai, Guangdong 519088, China (e-mail: xiejiangweiouc@gmail.com; mengmengzhang@bit.edu.cn; liwei089@ieee.org).

Yuxiang Zhang is with the Jockey Club STEM Laboratory of Quantitative Remote Sensing, Department of Geography, The University of Hong Kong, Hong Kong, China (e-mail: yxzhang7@hku.hk).

Yan Du is with the Communication Engineering Department, DFH Satellite Company Ltd., Beijing 100094, China (e-mail: dydhslgz@163.com).

Digital Object Identifier 10.1109/TGRS.2026.3713303

I. Introduction

SEMANTIC change detection (SCD) aims to identify both the locations of changes and the corresponding semantic categories of objects at the pixel level [1]. As a fundamental task in Earth observation, SCD supports numerous applications, including urban planning [2], disaster assessment [3], and environmental monitoring [4]. With the rapid growth of satellite imagery and Earth observation platforms, large-scale, high-resolution data are now widely available. While this enables more detailed monitoring of land-cover and urban dynamics, it also introduces several challenges for SCD. High spatial and spectral resolutions increase the complexity of feature extraction, and accurate semantic labeling often relies on labor-intensive manual annotation. Moreover, the temporal dimension of satellite image time series (SITS) adds further difficulty, as changes can be gradual, irregular, or subtle, making robust detection and classification more challenging.

To address SCD challenges, many approaches have been proposed. Early methods mainly relied on traditional image processing and statistical techniques, including [5] and support vector machines (SVMs) [6]. With rapid deep learning advances, learning-based methods [7], [8], [9] have become the dominant paradigm. Convolutional neural networks (CNNs) and later Transformer-based architectures [10] are widely used to automatically extract spatial features. Bandara and Patel [11] and Chen et al. [12] adopt Siamese architectures to replace traditional fully convolutional networks (FCNs). In learning strategy, multitask learning has also been explored [1], [8]. More recently, some methods transfer pretrained foundation models (FMs) to SCD via fine-tuning [13], while others train remote sensing-oriented FMs from scratch [14], [15]. These advances have consistently improved SCD performance.

Although conventional SCD has been widely studied, most approaches remain limited to bitemporal inputs, similar to binary change (BC) detection [16]. This formulation limits capturing gradual or complex land-cover dynamics [17]. In contrast, SITS provide richer temporal information, enabling continuous monitoring of urban and land-cover changes due to wide spatial coverage and frequent observations [18]. However, exploiting SITS for SCD brings new challenges. Models must process high-dimensional spatial–temporal data, handle irregular sampling, and remain robust to noise and atmospheric effects such as cloud coverage [19]. Motivated

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

by these advantages and advances in time series semantic segmentation [20], the SITS-SCD task has been proposed, where input is a sequence of multitemporal images rather than a bitemporal pair.

Early SITS-SCD methods typically rely on two-stage strategies, such as preclassification [19] or postclassification [21], where change detection and semantic labeling are performed separately. Moving beyond these two-stage algorithms, He et al. [18] proposed an end-to-end framework that directly learns the mapping between spectral features and land-cover categories, thereby predicting SCD maps in a unified manner. Nevertheless, this framework essentially treats the temporal sequence as an independent axis and collapses the spatial dimension into a single vector representation. Such a design overlooks rich spatial–spectral structures and weakens the ability to capture spatio-temporal interactions. As a result, its effectiveness largely depends on sufficiently long temporal sequences, while its applicability to short-term or irregularly sampled scenarios remains limited. Hafner et al. [22] proposed to utilize an attention mechanism along the sequence dimension to refine temporal embeddings, but it still struggles with spatial–spectral feature extraction, as it relies solely on a vanilla U-Net [23].

Despite these advances, existing SITS-SCD approaches suffer from three critical limitations that hinder their practical deployment: 1) they overemphasize temporal modeling while neglecting explicit spatial–temporal interaction, leaving rich spatial–spectral structures underexploited; (***2***) feature fusion remains shallow and inefficient—multiscale representations are either independently processed or naively concatenated, failing to mutually reinforce spatial and temporal cues during decoding; and 3) the robustness of these models under spatial domain shifts, where training and testing regions are geographically misaligned, remains largely unexplored, despite being a common scenario in real-world Earth observation.

To overcome these limitations, we propose a spatial–temporal collaborative network for the SITS-SCD task named ST3DUNet, which is applicable to both land use and land cover (LULC) and building detection scenarios, and supports both optical and multispectral modalities. To enable explicit spatial–temporal interaction, we design a bidirectional spatial–temporal module (BSTM) that jointly captures long-range dependencies across both space and time, rather than treating the temporal sequence as an isolated axis. To achieve effective feature fusion, we introduce a collaborative fusion block (CFB) that recursively integrates hierarchical representations at multiple scales via dynamically updated prototype matrices, guiding the interaction of spatial and temporal cues to enhance decoding quality. To evaluate robustness under domain shifts, we conduct comprehensive experiments under both in-domain and out-of-domain settings, providing insights into the generalization capability of learning-based SITS-SCD methods. Technically, ST3DUNet leverages a series of 3-D convolutions to facilitate spatial and spectral feature encoding, and the BSTM extracts hierarchical spatial–temporal representations that are progressively fused by the CFB throughout the decoder.

In summary, our main contributions are as follows.

1) We propose ST3DUNet, a novel spatial–temporal collaborative network for SITS-SCD, which jointly models spatial, temporal, and spectral dependencies, and is applicable to diverse SCD scenarios, including LULC and building monitoring.
2) We design a BSTM to capture long-range dependencies across both space and time, and a CFB that leverages dynamically updated prototype matrices to effectively integrate multiscale representations and inject informative cues into the decoding process.
3) We conduct an in-depth analysis of spatial domain shift across different regions, which reflects real-world SITS-SCD scenarios where testing areas are often misaligned with training regions, and provide insights into the generalization capability of learning-based SITS-SCD methods.
4) We perform extensive experiments on three challenging SITS-SCD datasets. ST3DUNet consistently achieves state-of-the-art performance, outperforming existing baselines and even two recent Earth observation FMs.

The remainder of this article is organized as follows. Section II reviews related work on SITS-SCD methods. Section III presents the details of our proposed ST3DUNet. Section IV reports experimental results, and Section V concludes the study.

II. Related Work

In this section, we will review previous SITS-SCD methods, which can be roughly divided into three categories: independent temporal processing, pairwise temporal modeling, and sequence-level temporal modeling. Each category will be discussed separately.

A. Independent Temporal Processing SITS-SCD

Independent temporal processing methods process each image independently at each timestamp. These methods first generate semantic segmentation maps and then perform pixel-level comparisons, where changes are identified based on discrepancies between the generated maps. In general, any semantic segmentation method can be applied to SITS-SCD. Early CNN-based approaches employed multilayer CNN encoders and FCN decoders [24], [25], [26]. Aerial LaneNe [25] incorporates wavelet transforms to improve cost-awareness within a symmetric encoder–decoder architecture. WiCoNet [27] adopts two CNN branches to separately capture global and local image-level features, thereby enhancing segmentation performance. Diakogiannis et al. [28] proposed a U-Net-style architecture with residual connections to improve feature extraction and contextual representation. To address the limitations imposed by local receptive fields, numerous attention-based methods and hybrid CNN–attention models have been developed. Li et al. [29] introduced a framework that applies spatial attention in shallow layers and channel attention in deep layers for satellite image semantic segmentation. Wang et al. [30] presented UNetFormer, a U-Net-like transformer architecture designed for efficient

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

semantic segmentation of remote sensing urban-scene imagery. Li et al. [31] proposed a hybrid framework that integrates CNN and transformer features with geometric priors to improve the semantic segmentation of high-resolution remote sensing images. To mitigate the quadratic computational complexity of transformers, Mamba-based models have recently gained increasing attention [32], [33], [34]. RS3Mamba [33] employs visual state-space (VSS) blocks to capture global contextual information and integrates it with local CNN features through a collaborative completion module. Xiao et al. [35] introduced an eight-directional selective scanning mechanism and multikernel parallel convolutions to capture rich global and local contextual information, while enhancing multiscale feature fusion through channel–spatial attention and atrous convolutions. Recently, FM-based methods [36], [37], [38] have demonstrated strong generalization capability and have been widely adopted as backbone models for a variety of downstream tasks. Liu et al. [39] proposed RemoteCLIP, a multimodal model trained with contrastive learning. RobSense [40] is designed to handle both single-modal and multimodal data, as well as static and temporal scenarios with missing frames. Alliance [41] was among the first to incorporate frequency-domain information, resulting in a unified spectral–spatial–frequency framework for remote sensing analysis.

B. Pairwise Temporal Modeling SITS-SCD

Pairwise temporal modeling methods typically extend conventional SCD approaches by taking two nearby frames from the time series, usually up to the last time point, as input. These methods generate two semantic segmentation maps and subsequently derive the SCD maps by comparing them.

Early pairwise temporal modeling methods commonly derive change features through simple differencing between temporal images [5]. However, such strategies often lose rich high-frequency information. Subsequent methods introduced Siamese networks with weight-sharing backbones to better model paired images. For example, Chen et al. [12] first employed a Transformer architecture for change detection, and Bandara and Patel [11] further developed this line of work by reducing the computational cost. Ding et al. [7] proposed a joint spatial–temporal modeling framework for SCD.

Beyond network design, multitask learning strategies have also been incorporated into pairwise temporal modeling SCD. Zheng et al. [1] were the first to introduce semantic segmentation into the SCD task, formulating it as a multitask learning problem. Tian et al. [8] exploited temporal consistency by designing a plug-and-play module to enhance change detection performance.

Recently, FMs have also been transferred to pairwise temporal modeling settings, typically within Siamese frameworks as feature encoders. For instance, Mei et al. [42] adapted the segment anything model (SAM) for SCD by incorporating task-specific modifications. Chen et al. [43] further introduced a "time traveling" mechanism to filter out irrelevant changes and focus on meaningful ones within Siamese networks.

C. Sequence-Level Temporal Modeling SITS-SCD

Several studies explicitly model temporal dependencies from sequential inputs. Rustowicz et al. [44] applied a 3-D U-Net for crop-type segmentation, while Garnot and Landrieu [45] introduced a U-Net with temporal attention for time series semantic and panoptic segmentation. Tarasiou et al. [46] adapted the Vision Transformer for sequential inputs in classification and segmentation. SkySense [47] utilized multigranularity contrastive learning within a teacher–student framework for large-scale pretraining, and incorporated a multimodal temporal fusion transformer as the neck to fuse temporal sequences. SkySense++ [48] further extended SkySense by enhancing its semantic understanding capability. Through masked semantic learning during pretraining, SkySense++ is able to better capture semantic information. Prithvi-EO-2.0 [49] is pretrained using masked autoencoder (MAE) techniques, where a 3-D convolution first processes sequence inputs, followed by transformer encoders to model temporal dependencies. However, these methods are primarily designed for time series land-cover classification or segmentation, typically assuming that semantic labels remain constant over time. As a result, although they capture temporal dynamics, they tend to aggregate the temporal dimension into a single representation and produce only one segmentation map, making them unsuitable for directly addressing SITS-SCD tasks. Therefore, we remove the temporal aggregation modules to preserve the sequence dimension, which consequently transforms these models into the independent temporal processing category.

Early sequence-level temporal modeling SITS-SCD methods adopted a two-stage pipeline, performing BC detection followed by pre- or postclassification [19], [21]. For example, Jing et al. [19] used differential operations with trajectory classification to monitor urban land-cover changes under heavy cloud conditions, while Zhu et al. [21] applied LASSO regression to detect land disturbance. Although related to SITS-SCD, these works focused on other objectives. More recently, He et al. [18] proposed an end-to-end FCN-based framework leveraging spectral changes, and Hafner et al. [22] incorporated temporal self-attention with CRF postprocessing to refine segmentation.

Despite these efforts, existing sequence-level temporal modeling methods often overemphasize temporal modeling while underexploring explicit spatial–temporal interaction. Consequently, rich spatial–temporal cues are not fully fused, and their potential for improving decoding is rarely exploited, limiting the effective use of complementary spatial and temporal information.

III. Method

As SITS-SCD is an emerging and rapidly evolving problem, we begin by presenting a clear problem statement to establish the foundation of our study.

A. Problem Statement

Given input SITS input data $\boldsymbol{X} = \{\boldsymbol{x}_i\}_{1:T} \in \mathbb{R}^{T\times C\times H\times W}$, where T is the time sequence length, C is the band number

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

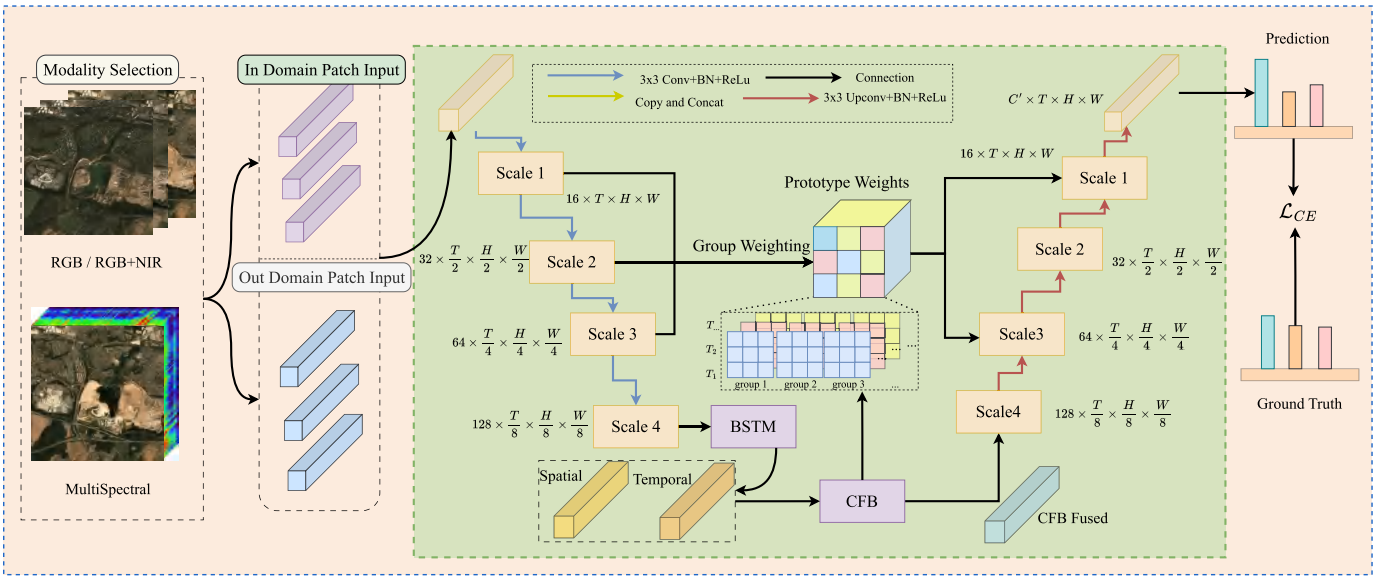


Fig. 1. Overview of ST3DUNet. Our framework adopts a U-shaped architecture and enhances the bottleneck stage by introducing BSTM and CFB modules beyond the standard encoder–decoder design, and the modeling of spatial–temporal collaborative information will also be injected into each scale, improving decoding quality.

and H, W is the spatial size, the objective of SITS-SCD is to output the semantic change (SC) map $\boldsymbol{Y} = \{\boldsymbol{y}_i\}_{1:M}$ between two time points, where M is the total change maps. When only adjacent change detection such as $(t_1, t_2), (t_2, t_3)$, is considered, $M = T - 1$.

For multitemporal SITS-SCD models, like our ST3DUNet, it outputs the semantic segmentation maps at each time stamp $\boldsymbol{S} = \{\boldsymbol{s}_i\}_{1:T} \in \mathbb{R}^{T\times H\times W}$ and conducts a per-pixel comparison to generate SC maps, which can be considered as a post classification comparison [8] category.

B. Overview

The overall architecture of ST3DUNet follows a U-shaped design and is illustrated in Fig. 1. Given an input sequence $\boldsymbol{X}$, we first apply a 3×3 3-D convolution, followed by batch normalization (BN) and a ReLU activation function, to extract an initial coarse feature $\boldsymbol{F}_0 \in \mathbb{R}^{16\times T\times H\times W}$. At each subsequent scale, a 2×2 max-pooling operation is employed, together with the same 3-D convolution–BN–ReLU block. The resulting intermediate feature at scale i doubles the channel dimension while halving both the temporal and spatial resolutions, i.e., $\boldsymbol{F}_0 \in \mathbb{R}^{16\times T\times H\times W}$, $\boldsymbol{F}_i \in \mathbb{R}^{2^i 16\times T/2^i\times H/2^i\times W/2^i}$.

At the bottleneck, the deepest feature is processed by the BSTM to enhance spatial–temporal feature representation. The resulting dual-branch spatial and temporal features are then effectively fused by the CFB, which outputs both the fused latent representation and corresponding weighting coefficients, consisting of a group of prototype matrices for other scales. The fused features are passed through the decoder, where spatial upsampling by a factor of 2 is performed at each stage, while the channel dimension is halved accordingly. Meanwhile, the weighting coefficients are upsampled and applied to the features $\boldsymbol{F}_i$, thereby implicitly enforcing spatial–temporal consistency during the decoding process. Finally, a 3-D convolution projects the decoded features into the semantic space, producing the final prediction maps.

C. Bidirectional Spatial–Temporal Module

The proposed BSTM is designed to jointly capture global and local temporal dependencies in SITS data while maintaining computational efficiency. As illustrated in Fig. 2, it follows a dual-branch design.

Let the input feature map be $\boldsymbol{F} \in \mathbb{R}^{T'\times H'W'\times C'}$, where T' denotes the temporal length, H' and W' are the spatial resolutions, and C' is the feature channel dimension after encoding.

To effectively capture long-range temporal dependencies, we design a lightweight temporal attention mechanism inspired by L-TAE [50]. This module adaptively re-weights the contribution of each time step, allowing the network to focus on temporally informative observations and model global temporal correlations.

Concretely, we first reshape the spatio-temporal feature tensor $\boldsymbol{F}$ into a 2-D representation $\tilde{\boldsymbol{F}} \in \mathbb{R}^{T'\times D}$, where $D = H' \times W' \times C'$ denotes the flattened spatial–spectral embedding dimension. A shared linear projection is then employed to generate the key and value representations through $\boldsymbol{W}_{KV} \in \mathbb{R}^{D\times D}$, ensuring parameter efficiency and temporal consistency.

To better capture high-level temporal semantics, we introduce a learnable global query vector $\boldsymbol{Q}_{\text{global}} \in \mathbb{R}^{1\times D}$, which serves as a content-agnostic temporal aggregator. Rather than attending to every frame equally, this query dynamically attends to the most informative time steps and integrates their contextual cues into a single coherent global representation. To retain the sequential structure of observations, temporal positional encoding TPE $\in \mathbb{R}^{T'\times D}$ is injected into the key representations derived from the input sequence, allowing the model to be aware of the temporal ordering of each frame during attention computation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

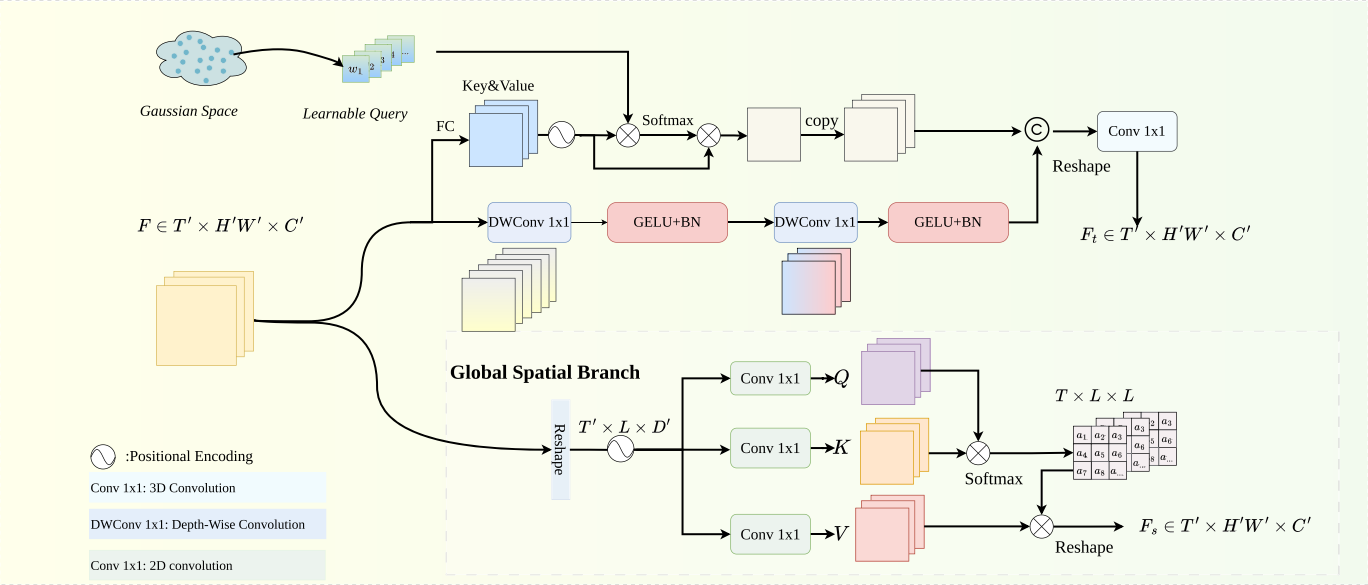


Fig. 2. BSTM consists of a temporal branch and a spatial branch for joint spatial–temporal modeling. The temporal branch combines a global L-TAE for long-range dependency modeling and a local temporal FFN with depth-wise 1-D convolutions for short-term dynamics. The spatial branch employs self-attention with spatial positional encoding across all time steps to capture global spatial relationships. The two branches operate in parallel to produce complementary spatial and temporal representations for subsequent fusion.

By decoupling the query from the input sequence itself, this formulation encourages the model to learn temporal priors that complement data-driven correlations, leading to more robust and interpretable temporal reasoning. The aggregated representation is subsequently broadcast across all T' time steps to produce the global temporal feature $\boldsymbol{F}_{\text{global}}$. The whole process can be formulated as follows:

$$\boldsymbol{K}_{\text{global}}, \boldsymbol{V}_{\text{global}} = \left(\tilde{\boldsymbol{F}} + \text{TPE}\right) \boldsymbol{W}_{\boldsymbol{KV}} \tag{1}$$

$$\text{Attn}_{\text{global}} = \text{Softmax}\left(\frac{\boldsymbol{Q}_{\text{global}} \boldsymbol{K}_{\text{global}}^{\top}}{\sqrt{D}}\right) \boldsymbol{V}_{\text{global}} \tag{2}$$

$$\boldsymbol{F}_{\text{global}} = \text{Repeat}\left(\text{Attn}_{\text{global}}, T'\right) \tag{3}$$

where $\text{Repeat}(\cdot, T')$ denotes duplication along the temporal dimension for T' times, and $\text{TPE} \in \mathbb{R}^{T' \times D}$ denotes temporal positional encoding.

To further capture fine-grained short-term variations, in parallel, we introduce a local temporal feed-forward network (FFN) with an expansion coefficient τ. Within this FFN, standard convolution layers are replaced by depth-wise 1×1 1-D convolutions, which operate as grouped convolutions along the temporal dimension. This design provides a balance between local temporal feature extraction and computational efficiency. Finally, the global and local temporal features are concatenated along the channel dimension and projected back to the input channel size via a 1×1 convolution, producing the temporal feature representation $\boldsymbol{F}_t \in \mathbb{R}^{T' \times H' W' \times C'}$, which can be expressed as follows:

$$\boldsymbol{F}_{\text{local}} = \sigma\left(\text{DWConv}\left(\sigma\left(\text{DWConv}\left(\boldsymbol{F}\right)\right)\right)\right) \tag{4}$$

$$\boldsymbol{F}_t = \text{Conv}_{1\times 1}\left(\left[\boldsymbol{F}_{\text{global}} \,\|\, \boldsymbol{F}_{\text{local}}\right]\right) \tag{5}$$

where $\sigma(\cdot)$ represents the GeLU activation followed by BN, and $\|$ indicates concatenation along the channel dimension.

For the spatial branch, we first divide the input feature map $\boldsymbol{F} \in \mathbb{R}^{T' \times H' W' \times C'}$ into patches, yielding token representations $\boldsymbol{Z}(t) \in \mathbb{R}^{L \times D'}$ at each time step t, where $L = (H'/p) \times (W'/p)$ is the total number of tokens, $D' = p \times p \times C'$ is the token feature dimension, and p denotes the patch size. Spatial positional encoding $\text{SPE}(t) \in \mathbb{R}^{L \times D'}$ is then added to obtain position-aware tokens $\tilde{\boldsymbol{Z}}(t) \in \mathbb{R}^{L \times D'}$. Three linear projections are applied to compute queries, keys, and values, followed by scaled dot-product attention to produce the spatial feature $\boldsymbol{F}_s \in \mathbb{R}^{T' \times H' W' \times C'}$. The process is formulated as follows:

$$\tilde{\boldsymbol{Z}}(t) = \boldsymbol{Z}(t) + \text{SPE}(t) \tag{6}$$

$$\boldsymbol{Q}(t) = \tilde{\boldsymbol{Z}}(t) \boldsymbol{W}_{\boldsymbol{Q}}(t) \tag{7}$$

$$\boldsymbol{K}(t) = \tilde{\boldsymbol{Z}}(t) \boldsymbol{W}_{\boldsymbol{K}}(t) \tag{8}$$

$$\boldsymbol{V}(t) = \tilde{\boldsymbol{Z}}(t) \boldsymbol{W}_{\boldsymbol{V}}(t) \tag{9}$$

$$\boldsymbol{F}_s(t) = \text{Softmax}\left(\frac{\boldsymbol{Q}(t)\, \boldsymbol{K}^{\top}(t)}{\sqrt{D'}}\right) \boldsymbol{V}(t) \tag{10}$$

where the projection matrices are defined as $\boldsymbol{W}_{\boldsymbol{Q}}(t), \boldsymbol{W}_{\boldsymbol{K}}(t), \boldsymbol{W}_{\boldsymbol{V}}(t) \in \mathbb{R}^{D' \times D'}$.

Note that, compared to the conventional self-attention mechanism, which ignores the temporal dimension, the affinity matrix $\boldsymbol{A}$ considers pairwise relations between patches at each time point throughout the temporal sequence.

D. Collaborative Fusion Block

The CFB aims to effectively integrate the spatial and temporal information extracted by the BSTM, as illustrated in Fig. 3. Given the spatial feature $\boldsymbol{F}_s \in \mathbb{R}^{T' \times H' W' \times C'}$ and the temporal feature $\boldsymbol{F}_t \in \mathbb{R}^{T' \times H' W' \times C'}$, we first compute cross-attention for each pixel across the entire temporal sequence,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

where $\boldsymbol{F}_t$ serves as the query and $\boldsymbol{F}_s$ serves as the key and value. This process can be formulated as follows:

$$\text{Attn}(t) = \text{Softmax}\left(\frac{\boldsymbol{F}_t(t)\,\boldsymbol{F}_s^{\top}(t)}{\sqrt{C'}}\right)\boldsymbol{F}_s(t). \tag{11}$$

Different from patch-based attention, we directly compute attention at the pixel level with G groups. Moreover, we do not further compute cross-attention across all spatial positions at each time step, considering both the limited sequence length and computational efficiency. Finally, a residual connection followed by a layer normalization is applied to stabilize and normalize the fused features.

Next, the affinity weights obtained from (11) are further condensed into prototype representations that summarize the spatial interaction patterns for each temporal step and position. For each query position m, we average the group g attention scores over all key positions $j = 1, \ldots, L$, yielding a position-specific prototype weight $a_{t,m}^{g}$ that captures how strongly that location interacts with the global spatial context at time step t. This query-centric aggregation ensures that each spatial location retains its own unique prototype weight rather than a globally shared one, which is semantically consistent with the subsequent position-wise reweighting of encoder features. The prototype weight matrix $\boldsymbol{A}^{g} \in \mathbb{R}^{T' \times H' \times W'}$ is obtained as follows:

$$a_{t,m}^{g} = \frac{1}{L}\sum_{j=1}^{L}\left[\text{Softmax}\left(\frac{\boldsymbol{F}_t^{m}(t)\;\boldsymbol{F}_s^{j\top}(t)}{\sqrt{C'}}\right)\right]^{g} \tag{12}$$

$$\boldsymbol{A}^{g} = \left\{a_{t,m}^{g}\right\}_{t=1:T',\,m=1:L} \in \mathbb{R}^{T' \times H' \times W'} \tag{13}$$

where $\boldsymbol{F}_t^{m}(t)$ denotes the temporal feature at query position m and time step t, and $\boldsymbol{F}_s^{j}(t)$ denotes the spatial feature at key position j. The assembled prototype matrix $\boldsymbol{A}^{g}$ is subsequently upsampled via interpolation to match both the spatial and temporal resolutions of each encoder feature map $\boldsymbol{F}_i$.

This prototype captures the global contextual importance of each spatial–temporal token and is subsequently used to reweight the corresponding feature representations. We then upsample $\boldsymbol{A}^{g}$ via interpolation to match both the spatial and temporal resolutions of each feature map $\boldsymbol{F}_i$. The resulting weight matrix is used to re-weight the encoded features, thereby injecting spatial–temporal awareness into the decoding stage. This can be expressed as follows:

$$\boldsymbol{A}_i^{g}(t,h,w) = \text{Interpolate}\left(\boldsymbol{A}^{g}\left(t',h',w'\right)\right) \tag{14}$$

$$\hat{\boldsymbol{F}}_i = \sum_{g}^{G}\sum_{t}^{T_i}\sum_{h,w}^{H_i,W_i}\boldsymbol{A}_i^{g}(t,h,w)\odot\boldsymbol{F}_i(t,h,w) \tag{15}$$

where G denotes the total number of groups, T_i, H_i, W_i represent sequence length and spatial size at branch i, $\odot$ indicates the Hadamard product. The fused representation $\hat{\boldsymbol{F}}_i$ integrates both spatial and temporal interaction patterns, and is subsequently forwarded to the decoder for final prediction.

IV. Experiments

A. Datasets and Evaluation Metrics

We evaluate our method on three datasets: DynamicEarthNet-RGB (Dynamic-RGB) [53], DynamicEarthNet-Sentinel2 (Dynamic-S2) [53], and MUDs [54]. The first two are different modalities of the DynamicEarthNet dataset: the RGB version provides four bands (RGB + NIR), while the Sentinel-2 version provides 12 spectral bands. All datasets contain 24 monthly images and corresponding annotations for LULC/building detection.

For DynamicEarthNet-RGB, we follow the official train/validation/test split, using 55 areas of interests (AOIs) for training and ten AOIs for validation and testing. For DynamicEarthNet-S2, we discard AOIs with fewer than 24 months of observations, resulting in 47 AOIs for training. For the MUDs dataset, we follow the split protocol described in [55].

For evaluation, we adopt the metrics defined by Toker et al. [53], including BC, SC, and SC segmentation (SCS) for SCD, as well as mean intersection-over-union (mIoU) for segmentation performance.

B. Implementation Details

We implement our method using the PyTorch library. The model is trained for 250 epochs with the AdamW optimizer, starting from an initial learning rate of 1×10^{-4} and decayed to 1×10^{-6} following a cosine annealing schedule. Our loss function is a class-weighted cross-entropy function. For all datasets, we uniformly sample $T = 8$ time steps from the 24-month sequences of each area. The input images are randomly cropped into 128×128 patches, and data augmentation includes random horizontal/vertical flips and rotations. The expansion coefficient τ in the temporal FFN is set to 4. More details can be found in our open source code. In all experiments, we consider all adjacent SCD among sequences, leading to total change maps $M = T - 1$.

For the domain shift experiments (Fig. 4), we consider two evaluation settings.

1) *In-Domain:* The test set consists of patches from the same geographical tiles as the training set but from nonoverlapping regions.
2) *Out-of-Domain:* The test set consists of patches from completely different geographical areas.

It is important to note that although training and testing patches are randomly cropped, the test patches are strictly drawn from nonoverlapping locations, thereby preventing data leakage between training and testing sets.

C. Comparison With State-of-the-Art Methods

We compare our ST3DUNet with independent temporal processing methods: U-TAE [45], TSViT [46]; pairwise temporal modeling methods: A2Net [51], SCanNet [7]; and sequence-level temporal modeling method: TSSCD [18]. For independent temporal processing methods, we remove the components designed for time series input to guarantee training is available. In addition, we incorporate three FMs: SkySense [47], AnySat [52], and Prithvi-EO-2.0 [49]. For both FMs, we use the officially released pretrained weights and attach a U-Net style segmentation head. If the pretrained weights are compatible with multitemporal inputs, we adopt the sequence-level temporal modeling setting; otherwise, we

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

TABLE I

RESULTS FOR ALL THREE DATASETS. FOR SKYSENSE, STM(S2) MEANS USING THE OFFICIAL SENTINEL-2 BACKBONE IN A MULTITEMPORAL MANNER. RED AND BLUE REPRESENT THE BEST AND SECOND-BEST VALUES. "-" INDICATES NOT APPLICABLE. ITP, PTM, AND STM DENOTE T INDEPENDENT TEMPORAL PROCESSING, PAIRWISE TEMPORAL MODELING, AND SEQUENCE-LEVEL TEMPORAL MODELING, RESPECTIVELY

Dataset	Method	Strategy	In domain				Out domain				Params↓ (M)
			mIoU ↑	BC ↑	SC ↑	SCS ↑	mIoU ↑	BC ↑	SC ↑	SCS ↑	
Dynamic-RGB	U-TAE [1] [45]	ITP	53.15	23.93	38.85	31.39	40.43	17.02	21.87	19.44	1.08
	TSViT [1] [46]	ITP	45.53	5.34	22.79	14.07	32.23	4.04	15.36	9.70	1.72
	A2Net [51]	PTM	39.92	18.15	21.91	20.03	38.14	16.20	21.81	19.00	1.83
	SCanNet [7]	PTM	49.29	20.21	31.09	25.65	38.61	14.59	22.11	18.35	27.90
	TSSCD [18]	STM	57.26	34.44	43.49	38.96	36.32	13.52	17.94	15.73	6.11
	SkySense [47]	ITP	46.47	26.66	35.72	31.19	37.55	16.89	25.50	21.19	661.99
	AnySat [52]	ITP	51.60	25.98	36.09	31.04	19.45	2.82	7.46	5.14	127.77
	AnySat [52]	STM	54.51	28.76	42.12	35.44	28.19	5.35	16.48	10.92	127.77
	PrithviEO2.0 [49]	STM	57.48	36.05	42.58	39.32	38.86	14.41	23.20	18.81	307.36
	ST3DUNet	STM	62.94	39.80	48.10	43.95	40.98	17.45	22.39	19.92	1.65
Dynamic-S2	U-TAE [45]	ITP	52.87	24.02	43.16	33.59	38.52	14.88	27.15	21.01	1.08
	TSViT [46]	ITP	45.89	3.19	20.58	11.89	13.45	2.85	8.78	5.81	1.72
	A2Net [51]	PTM	42.02	17.53	33.70	25.61	32.86	16.85	28.32	22.59	1.83
	SCanNet [7]	PTM	38.06	21.72	30.55	26.14	37.55	14.73	23.99	19.36	27.90
	TSSCD [18]	STM	54.77	34.47	41.76	38.12	29.02	9.88	14.20	12.04	6.11
	SkySense [47]	ITP	49.92	25.74	38.21	31.97	33.55	15.69	28.83	22.26	661.99
	AnySat [52]	STM	53.67	29.05	42.17	35.61	26.06	9.44	15.90	12.67	127.77
	SkySense [47]	STM (S2)	56.31	38.79	41.12	39.96	38.08	15.96	24.53	20.25	989.77
	PrithviEO2.0 [49]	STM	54.44	28.45	44.99	36.72	37.21	15.84	27.60	21.72	307.36
	ST3DUNet	STM	59.21	39.24	43.16	41.20	38.67	14.76	23.25	19.01	1.65
Muds	U-TAE [45]	ITP	63.94	1.02	32.54	16.78	59.99	1.07	30.15	15.61	1.08
	TSViT [46]	ITP	52.23	0.19	17.47	8.83	51.36	0.59	13.42	7.01	1.72
	A2Net [51]	PTM	59.08	1.19	41.24	21.21	54.79	0.92	38.54	19.73	1.83
	SCanNet [7]	PTM	62.90	0.91	23.68	12.30	59.68	1.05	27.51	14.28	27.90
	TSSCD [18]	STM	61.61	1.41	18.08	9.74	52.30	5.58	12.57	9.07	6.11
	SkySense [47]	ITP	62.14	1.31	44.30	22.81	57.44	1.01	31.43	16.22	661.99
	AnySat [52]	STM	62.90	1.34	42.57	21.95	54.53	0	0	0	127.77
	PrithviEO2.0 [49]	STM	62.82	5.81	30.83	18.32	58.19	4.22	25.86	15.04	307.36
	ST3DUNet	STM	69.28	3.01	30.97	16.99	64.89	4.23	29.47	16.85	1.65

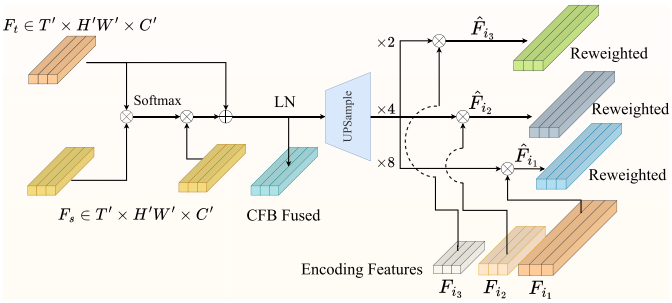


Fig. 3. Illustration of CFB. Given spatial and temporal features extracted by the BSTM, CFB first performs cross-attention to model spatial–temporal interactions. The resulting attention maps are condensed into prototype representations and used to reweight multiscale encoder features, enabling effective fusion of spatial and temporal information before decoding.

load the pretrained weights for corresponding channels and randomly initialize the remaining channels. Both FMs are fully fine-tuned with a learning rate of 5×10^{-6} until convergence.

1) DynamicEarthNet-RGB: The quantitative results are summarized in Table I. Our ST3DUNet consistently surpasses all baseline methods across all metrics under the in-domain setting, achieving an mIoU of 62.94. Moreover, it also establishes new SOTA performance under the out-of-domain condition, particularly on mIoU, demonstrating its strong generalization ability across unseen regions.

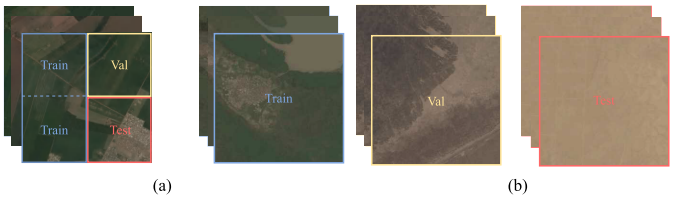


Fig. 4. Explanation of domain shift. No domain shift, i.e., in-domain, indicates that the validation and test sets are drawn from the same geographical area, while domain shift, i.e., out-of-domain, refers to sets from different geographical areas. (a) No domain shift. (b) Domain shift.

The qualitative comparisons are shown in Figs. 5 and 6. As illustrated in Fig. 5, our method produces segmentation maps with clearer object boundaries and more accurate class assignments compared to existing approaches. Fig. 6 further highlights the SCD results between two time points. While most baseline methods are capable of capturing BCs, they often struggle to correctly classify the changed categories. In contrast, our ST3DUNet provides predictions that are closer to the ground truth, benefiting from its explicit spatial–temporal modeling and collaborative fusion design. These results collectively confirm both the effectiveness and robustness of our approach for multitemporal remote sensing analysis.

2) DynamicEarthNet-Sentinel2: The quantitative results are summarized in Table I, which show a similar trend to those

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

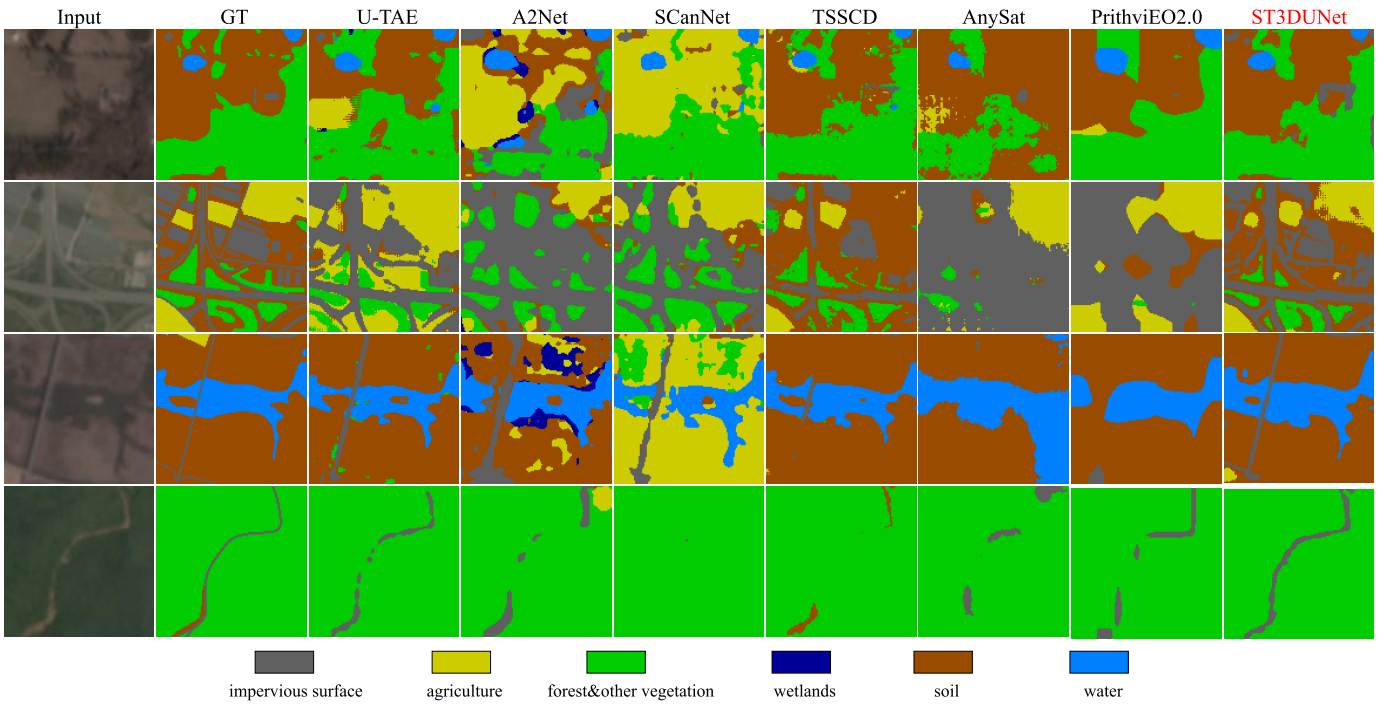


Fig. 5. Semantic segmentation results on the DynamicEarthNet-RGB dataset.

on the Dynamic-RGB dataset: our ST3DUNet consistently outperforms all baseline methods under the in-domain setting and achieves the best mIoU under the out-of-domain setting. Notably, there is a slight performance drop from Dynamic-RGB to Dynamic-S2. We attribute this to the relatively lower quality of the Dynamic-S2 dataset, such as cloud occlusion, as discussed in [53, Appendix].

The visualization results of semantic segmentation and SCD are presented in Figs. 7 and 8. These examples further demonstrate the robustness and effectiveness of our method across different LULC classes and from simple to complex scenarios, producing clearer segmentation maps and more accurate change boundaries compared to other methods. Moreover, the long sequence segmentation results in Fig. 8 show that our model outperforms other methods at all time stamps, except for a few time points, proving its robust segmentation ability.

3) Muds: The quantitative results are summarized in Table I. Our ST3DUNet achieves the best mIoU performance under both in-domain and out-of-domain settings, with scores of 69.28 and 64.89, respectively. Since the MUDs dataset contains only two classes, SCD here reduces to BC detection. In this case, where changes typically occupy small areas, mIoU primarily reflects the model's ability to precisely localize changes, while SC emphasizes semantic consistency. It is worth noting that SC may exaggerate the impact of small misclassifications in such binary settings, and thus should be interpreted in conjunction with both mIoU and qualitative results for a fair evaluation. It is worth noting that the AnySat model does not detect any changes under the domain shift condition, reflecting the unrobustness of transferring to domain gap scenarios.

The qualitative comparisons are shown in Figs. 9 and 10. As illustrated in Fig. 9, most baseline methods frequently confuse foreground and background regions, resulting in high false positives and false negatives, especially in complex scenarios. In contrast, our ST3DUNet produces cleaner boundaries and more reliable segmentation. Moreover, Fig. 10 highlights the SCD predictions between two time points: while the ground-truth change is nearly zero, only our method remains stable and correctly preserves the unchanged state, further demonstrating its robustness.

TABLE II
ABLATION STUDY RESULTS OF THE KEY COMPONENTS IN ST3DUNET. "IN" AND "OUT" INDICATE IN-DOMAIN AND OUT-OF-DOMAIN SETTINGS, RESPECTIVELY

Method	mIoU on different datasets			
	Dynamic-RGB (In)	Dynamic-RGB (Out)	Muds (In)	Muds (Out)
Basic Network	56.80	38.96	62.42	61.01
w/o CFB	58.10	39.22	66.13	63.00
w/o BSTM Spatial	59.62	39.30	67.17	63.05
w/o BSTM FFN	60.14	39.73	68.41	63.69
ST3DUNet	62.94	40.98	69.28	64.89

D. Ablation Study

In this section, we evaluate two aspects of ST3DUNet: key components and time sequence length. We conduct experiments on Dynamic-RGB and Muds datasets due to the relatively low data quality of Dynamic-S2 and report mIoU for simple.

1) Key Components: We designed five groups of experiments: 1) basic network, i.e., without CFB and BSTM; 2) without CFB; 3) without global spatial branch; 4) without local temporal FFN; and 5) full version of ST3DUNet. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

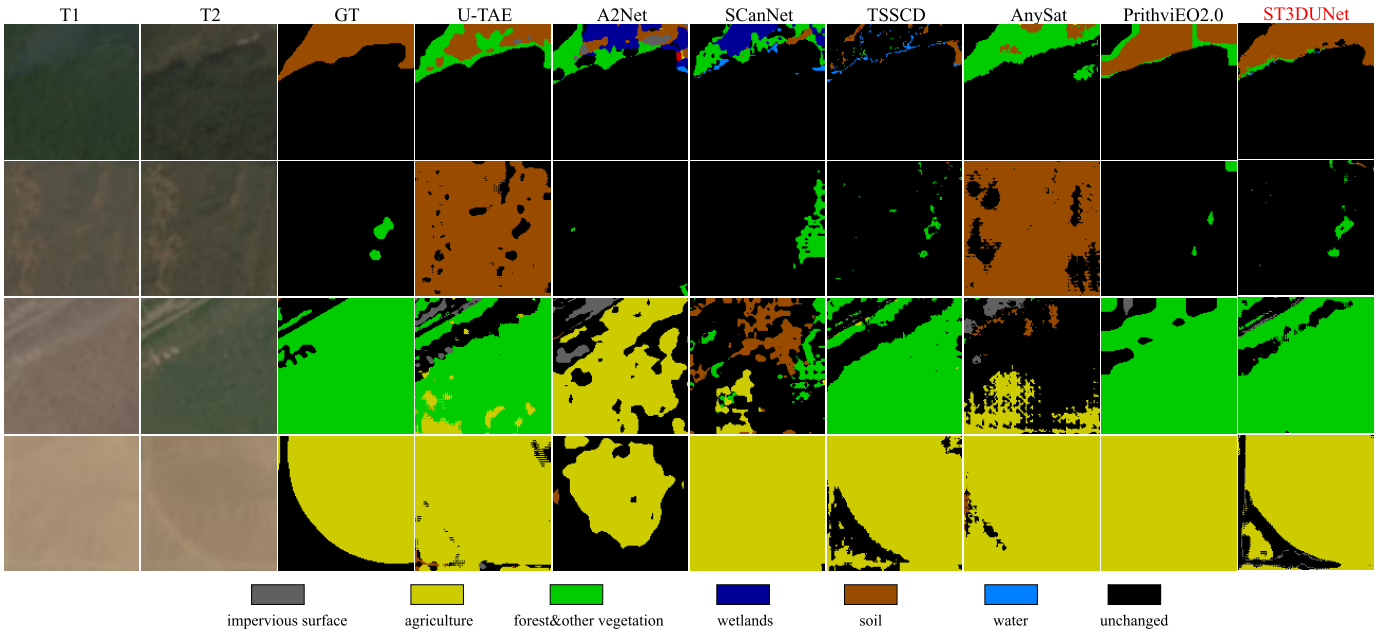


Fig. 6. SCD results on DynamicEarthNet-RGB dataset.

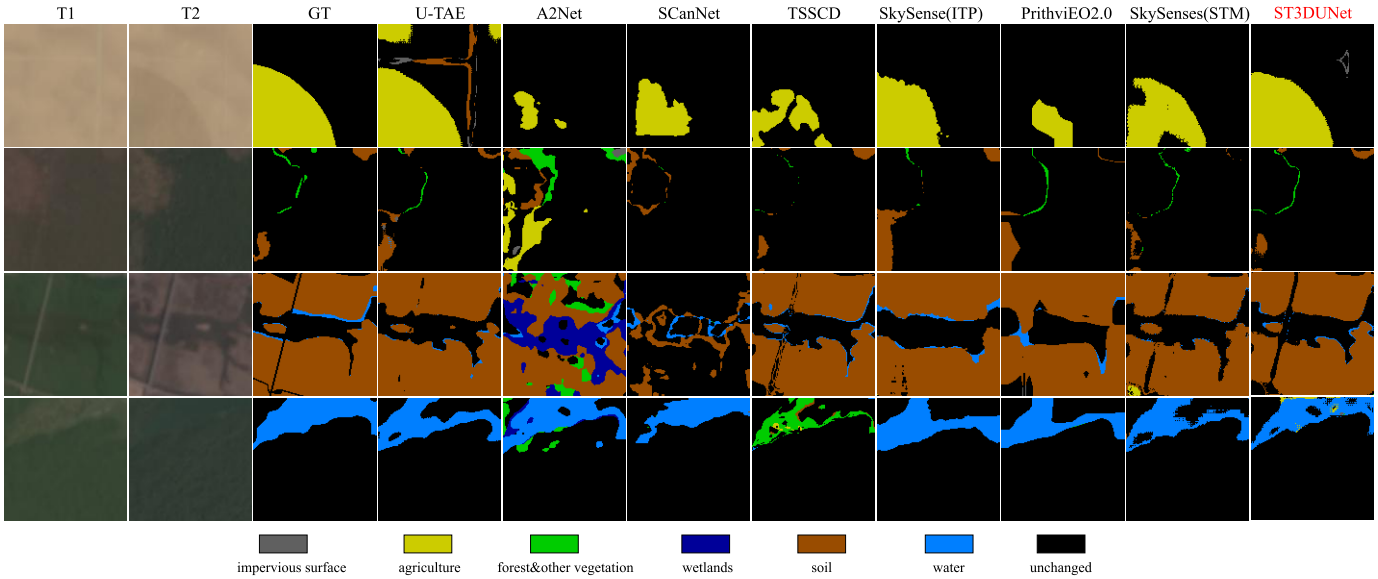


Fig. 7. SCD results on DynamicEarthNet-S2 dataset.

results are shown in Table II. It clearly shows the effectiveness of each component in our method; only when all components are incorporated, our ST3DUNet achieves the best performance.

2) Sequence Length: In our experiments, the sequence length is set to $T = 8$ for all datasets by default. To further investigate its impact, we additionally evaluate $T = 4$ and $T = 16$. When $T = 4$, the number of downsampling layers is reduced to three so that the bottleneck sequence length remains one-quarter of the input rather than one-eighth. The results are presented in Fig. 11. We observe that both $T = 8$ and $T = 16$ consistently outperform $T = 4$ across datasets, highlighting the benefit of leveraging richer temporal context. When $T = 16$, the performance is comparable to that with $T = 8$, occasionally surpassing or slightly falling behind it due to variations [56] introduced by longer sequences. Moreover, training with $T = 16$ requires significantly more computation and memory. Overall, $T = 8$ strikes the best balance between accuracy and efficiency, serving as an effective sweet spot in practice.

E. Time Stamp Sampling Mode

In our experiments, we primarily adopt a uniform sampling strategy, where T frames are evenly selected over 24 months. However, in practice, satellites often fail to provide regularly revisited observations due to weather conditions, cloud coverage, or acquisition gaps. To evaluate the robustness of our model under such conditions, we further consider

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

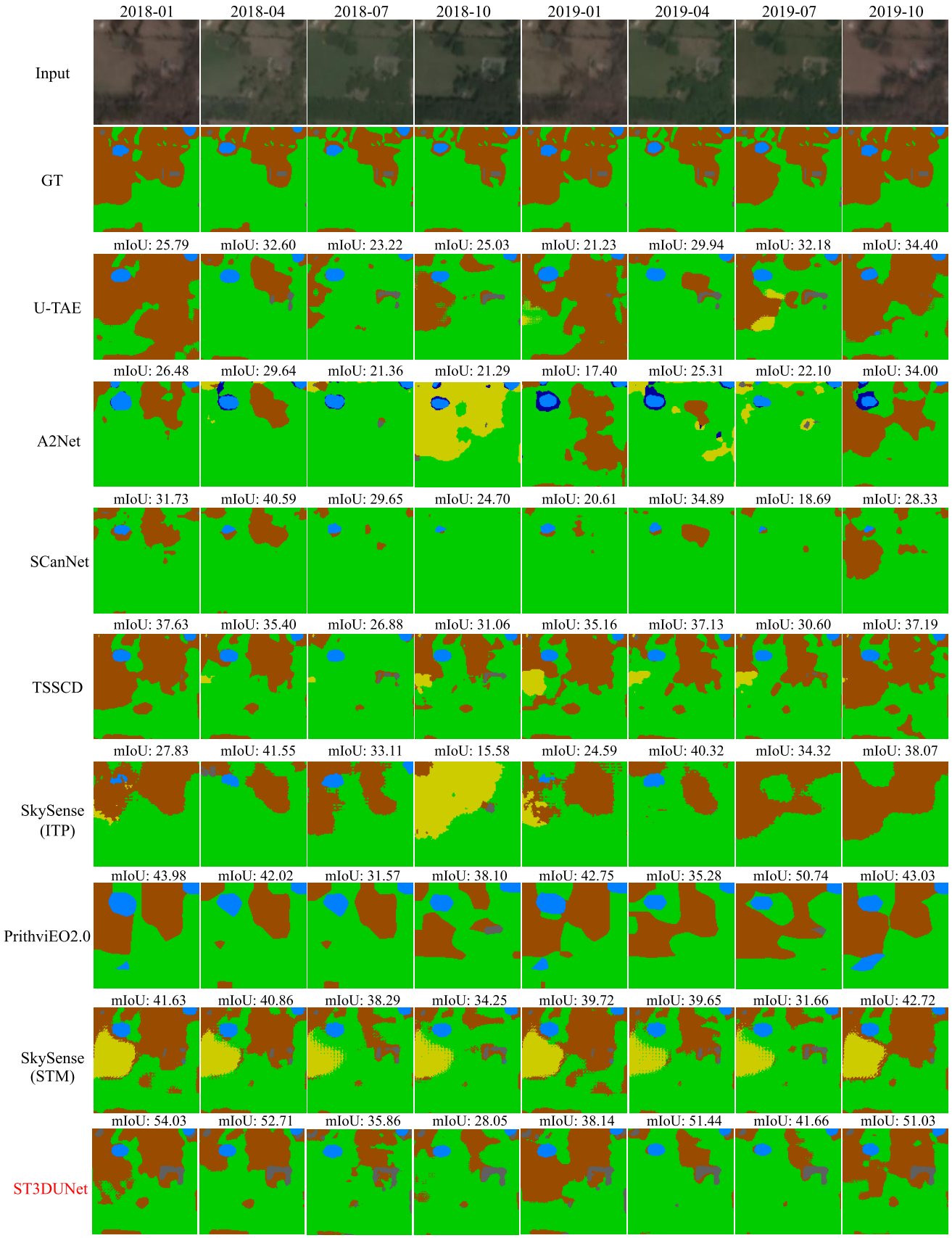


Fig. 8. Long sequence SCD results on the DynamicEarthNet-S2 dataset. We report mIoU above each image.

three alternative sampling modes: 1) random: randomly select T frames from the 24 months; 2) first: select the first T consecutive frames; and 3) last: select the last T consecutive frames. The random strategy mimics irregular revisit scenarios,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

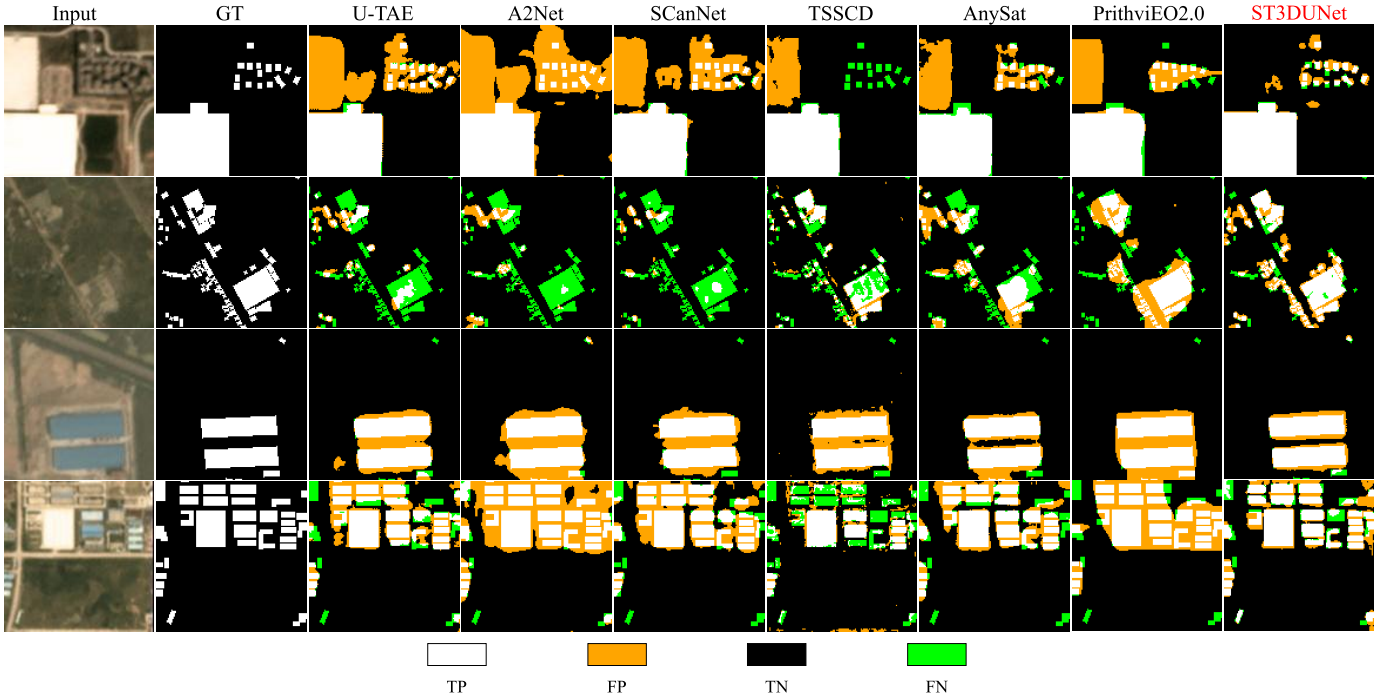


Fig. 9. Semantic segmentation results on the Muds dataset.

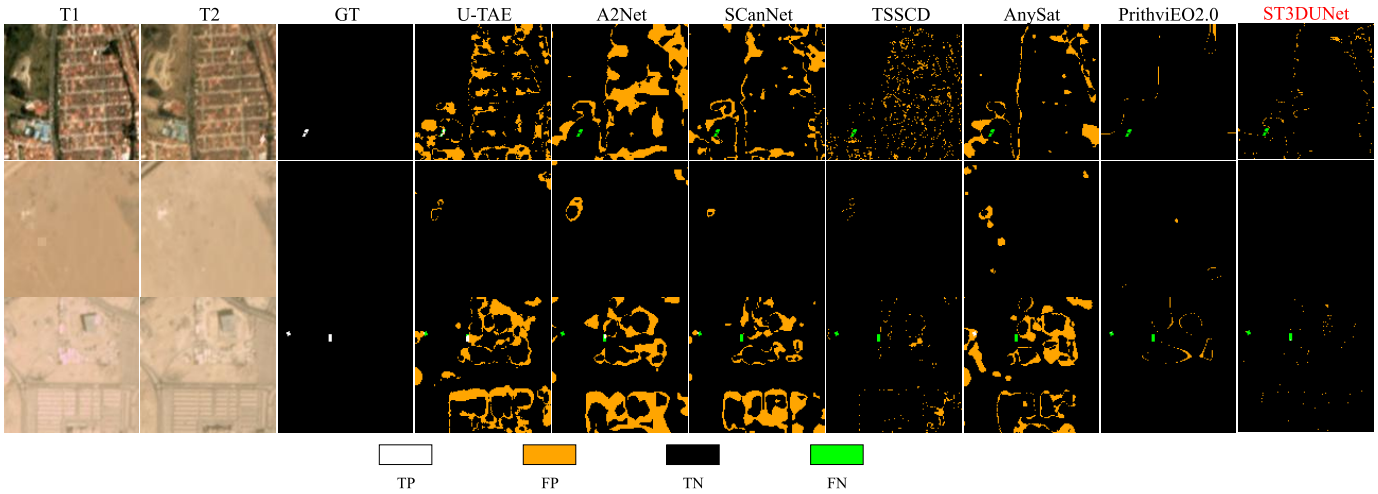


Fig. 10. SCD results on the Muds dataset.

TABLE III

RELATIONSHIP BETWEEN TIME STAMP SAMPLING MODES AND MIOU AND SCS METRICS. "IN" AND "OUT" DENOTE WITHOUT DOMAIN SHIFT AND DOMAIN SHIFT SETTINGS, RESPECTIVELY

Sampling	Dynamic-RGB(In)		Dynamic-RGB(Out)		Muds(In)		Muds(Out)	
	mIoU	SCS	mIoU	SCS	mIoU	SCS	mIoU	SCS
Random	55.54	32.63	43.22	19.26	67.51	15.31	64.54	16.38
First	61.83	36.34	41.53	15.37	70.03	17.31	64.42	11.35
Last	61.44	35.00	41.44	17.08	69.38	16.09	63.28	16.71
Uniform	62.94	43.95	40.98	19.92	69.28	16.99	64.89	16.85

while the first and last strategies correspond to high temporal resolution conditions with shorter revisit intervals. We conduct experiments on the Dynamic-RGB and Muds datasets with $T = 8$.

The results are summarized in Table III. Random sampling tends to benefit out-of-domain generalization by exposing the model to diverse temporal intervals, but it slightly compromises in-domain performance. In contrast, the high temporal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

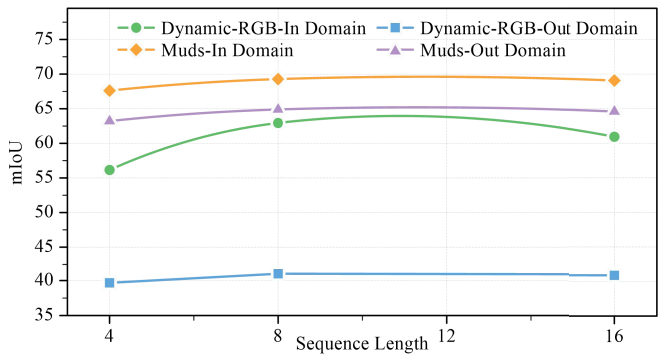


Fig. 11. Relationship between mIoU and sequence length.

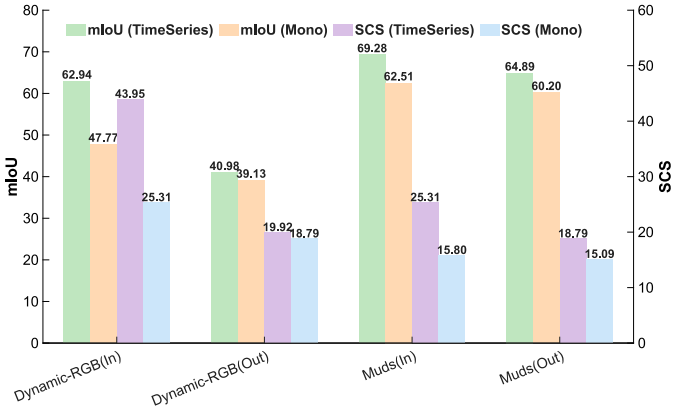


Fig. 12. Comparison between time series input and mono-temporal input. Our ST3DUNet benefits from time series.

resolution strategies (*First* and *Last*) yield similar results in the in-domain setting and show marginal improvements in the out-of-domain case, suggesting that shorter revisit intervals provide finer temporal cues compared to longer-interval sequences. Notably, almost all groups produce lower SCS scores than the uniform sampling baseline, indicating that a constant interval is generally preferable for change detection. On the other hand, higher temporal resolution benefits semantic segmentation. Overall, these findings highlight the importance of selecting an appropriate sampling strategy tailored to both the data characteristics and the target task.

F. Benefit From Time Series

In this section, we investigate the benefits of leveraging time series data compared to mono-temporal input. To this end, we construct a mono-temporal variant, ST2DUNet, by replacing all 3-D convolutions with 2-D convolutions and removing all temporal operations. Experiments are conducted on the Dynamic-RGB and MUDs datasets under both in-domain and out-of-domain settings, and both mIoU and SCS metrics are reported. The results, shown in Fig. 12, clearly demonstrate the performance gains achieved by incorporating temporal information. Specifically, compared to the mono-temporal version, ST3DUNet achieves mIoU improvements of 15.17 and 6.77 on in-domain settings, and 1.85 and 4.69 on out-of-domain settings for Dynamic-RGB and MUDs, respectively. These results highlight the importance of modeling temporal dependencies in SITS-SCD tasks.

V. Conclusion

In this article, we investigate the challenging task of SITS-SCD and propose a novel approach, ST3DUNet. Our method leverages BSTM to effectively capture both spatial and temporal informative representations, while the CFB introduces spatial–temporal relational modeling through a set of prototype-based matrices. Extensive experiments on three benchmark datasets validate the superiority and robustness of ST3DUNet. In future work, we plan to extend ST3DUNet to multimodal SITS-SCD scenarios, which would further enhance its applicability in real-world remote sensing tasks.

References

[1] Z. Zheng, Y. Zhong, S. Tian, A. Ma, and L. Zhang, "ChangeMask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 228–239, Jan. 2022.

[2] A. Javed, T. Kim, C. Lee, and Y. Han, "Deep-learning-based semantic change detection for urban greenery and comprehensive urban areas," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 1841–1852, 2025.

[3] J. Tu, D. Li, W. Feng, Q. Han, and H. Sui, "Detecting damaged building regions based on semantic scene change from multi-temporal high-resolution remote sensing images," *ISPRS Int. J. Geo-Inf.*, vol. 6, no. 5, p. 131, Apr. 2017.

[4] J. A. Cardille, E. Perez, M. A. Crowley, M. A. Wulder, J. C. White, and T. Hermosilla, "Multi-sensor change detection for within-year capture and labelling of forest disturbance," *Remote Sens. Environ.*, vol. 268, Jan. 2022, Art. no. 112741.

[5] T. Lei, Y. Zhang, Z. Lv, S. Li, S. Liu, and A. K. Nandi, "Landslide inventory mapping from bitemporal images using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 982–986, Jun. 2019.

[6] H. Nemmour and Y. Chibani, "Multiple support vector machines for land cover change detection: An application for mapping urban extensions," *ISPRS J. Photogramm. Remote Sens.*, vol. 61, no. 2, pp. 125–133, Nov. 2006.

[7] L. Ding, J. Zhang, H. Guo, K. Zhang, B. Liu, and L. Bruzzone, "Joint spatio-temporal modeling for semantic change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610814.

[8] S. Tian, A. Ma, Z. Zheng, X. Tan, and Y. Zhong, "Learning temporal consistency for high spatial resolution remote sensing imagery semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5622325.

[9] Y. Tang, S. Feng, C. Zhao, Y. Chen, Z. Lv, and W. Sun, "A semantic change detection network based on boundary detection and task interaction for high-resolution remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 9, pp. 17184–17198, Sep. 2025.

[10] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[11] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[12] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[13] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112.

[14] A. Fuller, K. Millard, and J. Green, "CROMA: Remote sensing representations with contrastive radar-optical masked autoencoders," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 5506–5538.

[15] M. Noman, M. Naseer, H. Cholakkal, R. M. Anwar, S. Khan, and F. S. Khan, "Rethinking transformers pre-training for multi-spectral satellite imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27811–27819.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.

[16] J. Xie, F. Gao, X. Zhou, and J. Dong, "Wavelet-based bi-dimensional aggregation network for SAR image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[17] Z. Zhu, "Change detection using Landsat time series: A review of frequencies, preprocessing, algorithms, and applications," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 370–384, Aug. 2017.

[18] H. He, J. Yan, D. Liang, Z. Sun, J. Li, and L. Wang, "Time-series land cover change detection using deep learning-based temporal semantic segmentation," *Remote Sens. Environ.*, vol. 305, May 2024, Art. no. 114101.

[19] C. Jing, W. Zhou, Y. Qian, W. Yu, and Z. Zheng, "A novel approach for quantifying high-frequency urban land cover changes at the block level with scarce clear-sky Landsat observations," *Remote Sens. Environ.*, vol. 255, Mar. 2021, Art. no. 112293.

[20] J. Yan, J. Liu, D. Liang, Y. Wang, J. Li, and L. Wang, "Semantic segmentation of land cover in urban areas by fusing multisource satellite image time series," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4410315.

[21] Z. Zhu et al., "Continuous monitoring of land disturbance based on Landsat time series," *Remote Sens. Environ.*, vol. 238, Mar. 2020, Art. no. 111116.

[22] S. Hafner, H. Fang, H. Azizpour, and Y. Ban, "Continuous urban change detection from satellite image time series with temporal feature refinement and multitask integration," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5627118.

[23] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[24] L. Mou and X. X. Zhu, "Vehicle instance segmentation from aerial image and video using a multitask learning residual fully convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6699–6711, Nov. 2018.

[25] S. M. Azimi, P. Fischer, M. Körner, and P. Reinartz, "Aerial LaneNet: Lane-marking semantic segmentation in aerial imagery using wavelet-enhanced cost-sensitive symmetric fully convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2920–2938, May 2019.

[26] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[27] L. Ding et al., "Looking outside the window: Wide-context transformer for the semantic segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410313.

[28] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet—A: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020.

[29] X. Li et al., "Dual attention deep fusion semantic segmentation networks of large-scale satellite remote-sensing images," *Int. J. Remote Sens.*, vol. 42, no. 9, pp. 3583–3610, May 2021.

[30] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[31] X. Li, F. Xu, F. Liu, Y. Tong, X. Lyu, and J. Zhou, "Semantic segmentation of remote sensing images by interactive representation refinement and geometric prior-guided inference," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5400318.

[32] E. Zhu, Z. Chen, D. Wang, H. Shi, X. Liu, and L. Wang, "UNetMamba: An efficient UNet-like mamba for semantic segmentation of high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[33] X. Ma, X. Zhang, and M.-O. Pun, "RS^{3}mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[34] M. Liu, J. Dan, Z. Lu, Y. Yu, Y. Li, and X. Li, "CM-UNet: Hybrid CNN-mamba UNet for remote sensing image semantic segmentation," 2024, *arXiv:2405.10530*.

[35] P. Xiao et al., "MF-mamba: Multiscale convolution and mamba fusion model for semantic segmentation of remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5405916.

[36] Y. Cong et al., "SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 197–211.

[37] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, 2022, Art. no. 5612822.

[38] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[39] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622216.

[40] M. K. Do, K. Han, P. Lai, K. T. Phan, and W. Xiang, "RobSense: A robust multi-modal foundation model for remote sensing with static, temporal, and incomplete data adaptability," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 7427–7436.

[41] B. Zhao et al., "Alliance: All-in-one spectral–spatial-frequency awareness foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 48, no. 3, pp. 3586–3603, Mar. 2026.

[42] L. Mei et al., "SCD-SAM: Adapting segment anything model for semantic change detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5626713.

[43] K. Chen et al., "Time travelling pixels: Bitemporal features integration with foundation model for remote sensing image change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 8581–8584.

[44] R. M. Rustowicz, R. Cheong, L. Wang, S. Ermon, M. Burke, and D. Lobell, "Semantic segmentation of crop type in Africa: A novel dataset and analysis of deep learning methods," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2019, pp. 75–82.

[45] V. S. Fare Garnot and L. Landrieu, "Panoptic segmentation of satellite image time series with convolutional temporal attention networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 4872–4881.

[46] M. Tarasiou, E. Chavez, and S. Zafeiriou, "ViTs for SITS: Vision transformers for satellite image time series," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10418–10428.

[47] X. Guo et al., "SkySense: A multi-modal remote sensing foundation model towards universal interpretation for Earth observation imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27662–27673.

[48] K. Wu et al., "A semantic-enhanced multi-modal remote sensing foundation model for Earth observation," *Nature Mach. Intell.*, vol. 7, no. 8, pp. 1235–1249, Aug. 2025.

[49] D. Szwarcman et al., "Prithvi-EO-2.0: A versatile multi-temporal foundation model for Earth observation applications," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2025, Art. no. 4400120.

[50] V. S. F. Garnot and L. Landrieu, "Lightweight temporal self-attention for classifying satellite images time series," in *Proc. Int. Workshop Adv. Anal. Learn. Temporal Data*, 2020, pp. 171–181.

[51] Z. Li et al., "Lightweight remote sensing change detection with progressive feature aggregation and supervised attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602812.

[52] G. Astruc, N. Gonthier, C. Mallet, and L. Landrieu, "AnySat: One Earth observation model for many resolutions, scales, and modalities," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 19530–19540.

[53] A. Toker et al., "DynamicEarthNet: Daily multi-spectral satellite dataset for semantic change segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 21126–21135.

[54] A. Van Etten, D. Hogan, J. M. Manso, J. Shermeyer, N. Weir, and R. Lewis, "The multi-temporal urban development SpaceNet dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6394–6403.

[55] E. Vincent, J. Ponce, and M. Aubry, "Satellite image time series semantic change detection: Novel architecture and analysis of domain shift," 2024, *arXiv:2407.07616*.

[56] J. Zhou, Z. Pang, and Y.-X. Wang, "RMem: Restricted memory banks improve video object segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 18602–18611.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:08:12 UTC from IEEE Xplore. Restrictions apply.